Spatial Correlation-Constrained Low-Rank Modeling for SAR Image Change Detection

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***Abstract*— Superpixel analysis is showing great potential for high-resolution synthetic aperture radar (SAR) image change detection (CD), as it uses a larger detection granularity and enhances computational efficiency. However, some deficiencies still exist. It is difficult for previous methods to extract the complete change regions from cluttered backgrounds. Meanwhile, the local spatial correlation and structure consistency are not well represented. To address the above problems and achieve better separation of changed and unchanged superpixels in complicated scenarios, we design a novel unsupervised CD framework from the perspective of low-rank matrix decomposition (LRMD) theory. The entire framework is carried out in two stages. First, the ℓ_1-norm sparsity constraint LRMD model is constructed to decompose change features into a low-rank component associated with background and a sparse component representing changed regions. Then, the local spatial correlation and structure consistency constraint are explicitly modeled by introducing a Laplacian regularization term. The unified model smooths the local similarity superpixels and enlarges the distance between changed regions and the background in the feature subspace. In this stage, the saliency difference image (DI) is generated to indicate the change probabilities of each superpixel. Furthermore, a classification refining (CR) module is designed to learn the projection from the change feature matrix to the saliency DI, which can further fine-tune such obscure regions and boost the binary classification. Extensive experiments on five challenging datasets from the TerraSAR-X sensor demonstrate the effectiveness and superiority of the proposed method.**

***Index Terms*— Change detection (CD), classification refining (CR), low-rank matrix decomposition (LRMD), synthetic aperture radar (SAR).**

NOMENCLATURE

$I = \{I_t\}_{1 \le t \le N}$	A set of SAR images.
$\mathbf{F} \in \mathbb{R}^{D \times N}$	Change feature matrix.
$\mathbf{L} \in \mathbb{R}^{D \times N}$	Low-rank matrix.
$\mathbf{S} \in \mathbb{R}^{D \times N}$	Sparse matrix.
$\mathbf{W} \in \mathbb{R}^{N \times N}$	Similarity matrix.
$\mathbf{D} \in \mathbb{R}^{N \times N}$	Degree matrix.
$\mathbf{M} \in \mathbb{R}^{N \times N}$	Laplacian matrix.
$\mathbf{X_i}$	ith column of matrix $\mathbf{X}$.
$\|\cdot\|_0$	ℓ_0-norm.
$\|\cdot\|_1$	ℓ_1-norm.
$\|\cdot\|_*$	Nuclear norm.
$\|\cdot\|_F$	Frobenius norm.
$G = (V, E)$	Graph with vertex set V and edge set E.
$\text{tr}(\mathbf{X})$	Trace of matrix $\mathbf{X}$.
$\mathbf{X^i}$	ith iteration of matrix $\mathbf{X}$.
$\mathbf{P} \in \mathbb{R}^{D \times 2}$	Projection matrix.
$\tilde{A}_i \in \mathbb{R}^{N \times 2}$	Sample matrix.
$\tilde{B}_i \in \mathbb{R}^{N \times 2}$	Label matrix.

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I. INTRODUCTION

CHANGE detection (CD) is one of the most widely used technologies in the field of remote-sensing Earth observation. It aims to highlight changed regions by utilizing bitemporal or multitemporal images over the same area at different dates. As an advanced active remote-sensing technology, synthetic aperture radar (SAR) works under nearly all weather conditions and at any time, which provides crucial data support for acquiring land change information in harsh environments. In recent years, SAR image CD has found broad applications in urban area monitoring [1], [2], [3], disaster assessment [4], [5], [6], environmental monitoring [7], [8], [9], and so on.

In the early stages of remote-sensing CD research, pixel-based CD (PBCD) methods have been the mainstream for a long time. These methods treat a pixel as the basic analysis unit and have been widely implemented in low- to medium-resolution SAR images. Most of them are paying more attention to extracting the changed areas from the background and distinguishing the changed boundaries as accurately as possible. The essence of the PBCD methods is to extract those changed pixels and filter those unchanged by computing the differences in backscattering between preevent and postevent images, which are extremely sensitive to speckle noise and clutter.

Benefiting from the rapid development of SAR satellite remote-sensing technology, a large amount of high-resolution data is easily available, which drives CD to transform from PBCD to object-based CD (OBCD) [10]. With the growth of spatial resolution, the increased variability within the changed areas and the higher complexity of change boundaries weaken the performance of PBCD methods. It often leads

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to a noisy output, with isolated changed pixels, holes in the middle of connected changed components, and jagged boundaries [11]. Therefore, the superpixel replaces the pixel to be the better unit for change analysis, as it can offer more useful spatial and structural information. Superpixel, namely an image object, is a collection of pixels that represents visually meaningful and locally similar regions [12]. For OBCD methods, superpixel segmentation is commonly used in the preprocessing step, and then these superpixels are directly compared and ascertained whether they have changed or not. Despite the advantages of suppressing noise and higher computation efficiency, superpixel-level OBCD methods have two major defects: 1) classification accuracy relies heavily on the initial segmentation results and 2) it is hard to separate the changed regions from the background when the differences in backscattering are slight. These limitations prevent their wide application in change monitoring.

In this article, we aim to propose an unsupervised low-rank matrix decomposition (LRMD) method for the OBCD problem. Specifically, we address the problem of OBCD for SAR images from the perspective of visual saliency computation. Inspired by recent research in salient object detection and the fact that changed regions are salient regions in difference images (DI), we introduce the popular LRMD framework to solve the OBCD problems mentioned above.

Traditional CD theory commonly assumes that only a small part of the observed scenes is changed, which can be considered to satisfy the approximate sparsity condition. Hence, the CD task can be transformed into an LRMD problem, in which the low-rank part can be seen as the unchanged areas and the sparse part corresponds to the changed areas [13]. During the decomposition process, there are two major concerns: 1) how to make full use of the relations between local superpixels and 2) how to refine the classes of those superpixels that have fuzziness difference values. Therefore, we propose a coarse-to-fine change information extraction framework to improve the performance from the above two points.

First, a main deficiency of the LRMD methods is that they ignore the intercorrelations between the elements in the sparse matrix, and thus the spatial relations between the superpixels are discarded, for example, spatial contiguity and local change consistency. It may cause inconsistent salient values between the superpixels within the same changed region. To address this problem, a Laplacian regularization is integrated into the ℓ_1-norm sparsity-constrained LRMD model to enhance the correlations of the elements in the sparse matrix. The whole model is based on a sensible local invariance assumption that the spatially adjacent superpixels with similar features should have similar representations in the subspace and thus have similar salient values. The regularizer has an important effect on partial smoothing, and the local structure information will be better preserved. As a result, the separability between the change regions and the background will be improved as much as possible. Meanwhile, more completely changed regions are consistently highlighted, and those isolated change points in the background will be suppressed.

Second, when the observation scenes are chaotic and the changed areas submerge in the background, it is hard for traditional LRMD methods to decompose the accurate sparse matrix. This will cause further misclassification in the final stage. To address this issue, a refinement module is constructed to learn the projection from the original change feature matrix to the saliency DI. A simple, but effective threshold approach is adopted to select positive (changed), negative (unchanged), and intermediate (uncertain) samples, and the pseudo-labels of those intermediate samples are determined by utilizing the local spatial information. The refinement process is completely unsupervised, and it does not need any labeled samples. In this step, the fuzzy superpixels will be assigned more accurate salient values, and the results of the final classification will be further optimized. In summary, the contribution of this work is threefold.

1) An ℓ_1-norm sparsity-constrained LRMD model is proposed for OBCD. To enhance the separability between the change regions and the background, a Laplacian regularization is introduced in the model. The alternating direction method of multipliers (ADMM) optimization algorithm is employed to solve the proposed LRMD model.
2) A classification refining (CR) module is constructed to optimize the salient values of those obscure superpixels. The process of refinement is completed by learning the saliency feature mapping, and it is transformed into an unsupervised joint optimization problem using both certain and uncertain samples.
3) The TerraSAR-X high-resolution data, acquired during the 2011 Sendai Tsunami in Japan, is exploited to verify the superiority and practicality of the proposed framework. Especially, except for bitemporal images, the proposed framework also shows better performance in multitemporal images.

The remainder of this article is organized as follows. Section II briefly reviews the related works on SAR image CD, especially the OBCD methods. Section III elaborates the proposed framework in detail. The extensive experimental results and detailed analysis are presented in Section IV. Finally, we conclude this article in Section V. For easier expression, we describe the notations used in the remainder of this article as shown in Nomenclature.

II. Related Work

Depending on the basic analysis unit, CD methods can be categorized into PBCD and OBCD [14]. The analysis granularity can be selected based on the resolution of the input data. In the following, we will present a detailed overview of the recent PBCD and OBCD methods.

A. PBCD Methods

For PBCD methods, individual pixels are treated as the basic analysis unit, and the main work is to categorize all pixels into changed or unchanged classes. Usually, the difference of the backscattering amplitude or intensity values between two pixels is used as the numerical basis for categorization. Among the most mature and widely used methods is the log ratio (LR) operator [15], [16], this was followed by

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some strategies such as image fusion [17], [18], [19], feature transforming [20], [21], [22], similarity measurement [23], [24], and deep learning [25], [26], [27], [28]. The methods indicated above are focused on the bitemporal images and are limited to representing change information in the time dimension. Recently, time-series analysis has blossomed due to the availability of tons of historical data. Time-series CD (TSCD) can not only change location and extent, but also obtain extra-temporal evolution information within the observation period [29]. A common strategy is to build a CD matrix (CDM) for each pixel by using pairwise traversal comparison, which records all possible changes in the time series [30], [31]. Statistical modeling is another effective method. Quin et al. [32] proposed a statistical-based TSCD method called MIMOSA. The key is to compute the theoretical joint probability density function between the geometric and quadratic means of the SAR image time series. In addition, the temporal coefficient of variation (CV) has been applied to detecting urban area changes [33]. Furthermore, changing temporal patterns are becoming a research focus. Su et al. [34] used a normalized cut-based clustering method to identify four main change patterns. In [35], the SAR-TST model was constructed to recognize the four change patterns of interest, that is, impulse, step, cycle, and complex.

B. OBCD Methods

In high-resolution SAR images, change regions usually present the characteristics of large-scale and complex spatial structures. The change intensity is inhomogeneous and fluctuant for those pixels within the change area. Therefore, it is difficult to extract the complete change regions with PBCD methods. OBCD methods offer ideal solutions to the above problems. The changed area can be regarded as one or several connected superpixels, namely image objects, which can suppress the noise false alarm and diminish the holes generated in the changed area.

Gong et al. [36] proposed a novel OBCD method that incorporates superpixel-based change feature extraction and hierarchical difference representation learning by DNN, which has been applied to optical image CD. In [37], a stacked contractive autoencoder (SCAE) was first presented to extract the temporal change features from the superpixels in DI. Another similar approach was proposed by Yu et al. [38], multiscale superpixel segmentation was adopted in the original SAR images, and then stacked denoising autoencoder (SDAE) was used to learn the difference representation between bitemporal image objects. In [39], a multiscale superpixel reconstruction method was developed to generate an optimized DI map. Then, a deep convolutional generative adversarial network (DCGAN) was designed to classify the changed and unchanged pixels. In this work, the superpixels were not used as the basic unit but for generating the initial DI. Therefore, it is still a pixel-level method. Sun et al. [40] proposed a structured graph-based image regression framework for multimodel CD tasks. The final binary change map was generated by the Markov segmentation model.

Besides, LRMD theory has been successfully applied to OBCD. Zhang et al. [41], first attempted to extract the sparse part using the GoDec algorithm from the original DI, which corresponded to the changed areas. Hou et al. [13] proposed a new method that incorporated low-rank saliency computation and deep feature representation. The changed part was extracted using the robust principal component analysis (RPCA) algorithm. In [42], low-rank representation (LRR) was employed to find the initial changed and unchanged superpixels and the PCA algorithm was used for training these samples. This method has been implemented in the multispectral image CD task. Ma et al. [43] proposed a matrix factorization method to obtain the changed statistical features, which were used to build a Bayes classifier to distinguish different pixels. Unfortunately, most of these methods simply extract sparse features through LRMD theory without any consideration for complicated scenes. The extracted features are either directly classified or used to support the subsequent pixel-based classification. In this article, we first argue that LRMD theory is more suitable for high-resolution SAR image CDs. Meanwhile, it should be more concerned that the constructed model is directly implemented on superpixels rather than aided by other pixel-based algorithms. The goal of developing OBCD methods is to maximize detection accuracy while maintaining computational efficiency. On this basis, the spatial correlation and structure consistency among OBCD analysis units should be fully taken into consideration owing to their higher spatial resolution. Therefore, we propose a novel LRMD model for the OBCD task that integrates the spatial correlation and change saliency information. It significantly improves classification accuracy.

III. Methodology

In this section, we will describe our OBCD algorithm that uses the proposed LRMD model, which consists of three steps: 1) preprocessing and superpixel segmentation; 2) saliency-changed area extraction; and 3) classification refinement. Fig. 1 shows the overall framework of the proposed method.

A. Preprocessing and Superpixel Segmentation

Multitemporal image preprocessing is the first step for the CD task. It mainly includes coregistration and filtering. The details of the preprocessing methods can be seen in the recently published article [35], and they will not be repeated here.

The next important step is superpixel segmentation. It groups the pixels by utilizing the similarity of features between the pixels and expresses the image features by replacing a large number of the pixels with a small number of superpixels. Thus, the complexity of the postprocessing step is largely reduced. As a classical algorithm, simple linear iterative clustering (SLIC) [44] has been widely used in SAR images for its advantages in both boundary consistency and calculation efficiency. The superpixels generated by SLIC are as compact and neat as cells, and the neighborhood features are easier to express. However, the presence of the speckle noise

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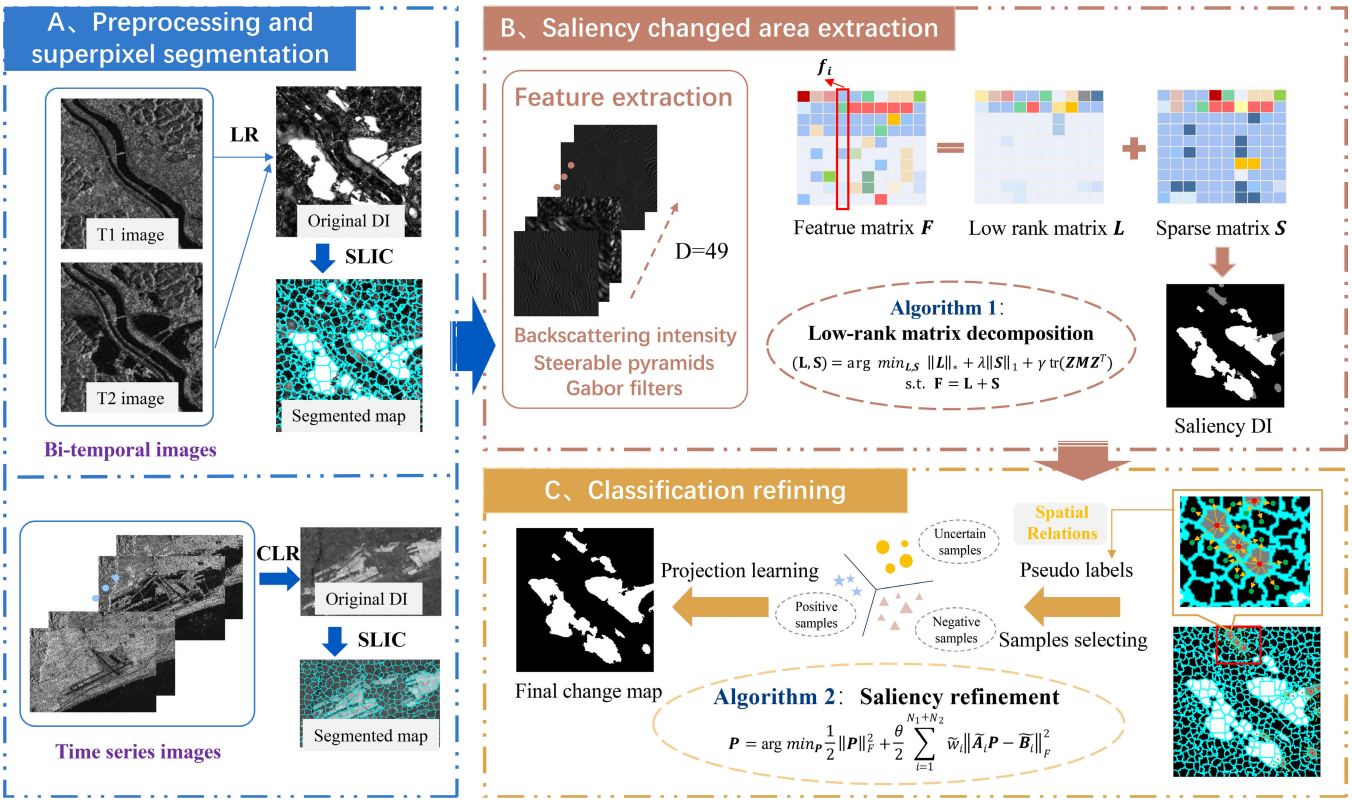


Fig. 1. Schematic of the proposed CD framework. It can be divided into three parts. Data preprocessing and superpixel segmentation are first conducted (module A), and then the Laplacian regularizing constrained LRMD model is constructed to extract saliency-changed areas (module B). Finally, the CR module is designed to learn a projection from raw features to saliency values, which achieves a refinement classification for those uncertain samples (module C).

degrades segmentation performance by directly applying SLIC to the original SAR images. To reduce segmentation errors, SLIC is conducted in the DI rather than the original images in the proposed framework. Specifically, let $I = \{I_t\}_{1 \le t \le N}$ be the set of SAR data, where $N \ge 2$. The LR operator is first used to generate the original DI. For bitemporal images, the standard LR operator is

$$\mathrm{DI}_{\mathrm{bi}}(x) = \log \left| \frac{I_1(x)}{I_2(x)} \right| \tag{1}$$

where $\mathrm{DI}(x)$ represents the difference value in position x. For multitemporal images, we define a new change indicator called the cumulative LR (CLR) operator. The expression is

$$\mathrm{DI}_{\mathrm{multi}}(x) = \mathrm{Norm}\left\{ \sum_{i=1}^{N} \log \frac{I_{i+1}(x)}{I_i(x)} \right\} \tag{2}$$

where $\mathrm{Norm}(\cdot)$ means the normalization operator. Theoretically, the temporal CV [33], [35] is a better change indicator for long time-series (dozens or even hundreds) data. However, when the number of historical images is very small (a few or a dozen), it will lead to large statistical errors. In this condition, CLR is better than CV to measure the change frequency in a short time period.

Once the segmentation map is obtained, the feature matrix can be constructed to apply the LRMD algorithm. For the CD task, the key to constructing the feature matrix is to extract representative change features. The backscattering intensity is the most essential feature for SAR images, thus the difference in the backscattering intensity is set to be the basic change feature. The steerable pyramid filters and the Gabor filters are both widely used feature extractors in low-rank modeling questions [45]. Both the texture and edge features can be better extracted using the above filters.

1) *Backscattering Intensity:* The difference in the backscattering intensity is set to be the one-dimensional feature.
2) *Steerable Pyramids:* Steerable pyramid filters are conducted on the DI with three layers and four directions, generating 12-D features.
3) *Gabor Filters:* Gabor filters are used to generate responses from three scales and 12 directions. The total 36-D features are extracted.

All the extracted 49-D features are then stacked vertically, and each superpixel is represented by a feature vector f_i. Finally, we construct the change feature matrix representation of the entire DI: $\mathbf{F} = [\mathbf{f_1}, \mathbf{f_1}, \ldots, \mathbf{f_N}] \in \mathbb{R}^{D \times N}$, where D is the dimension of the feature vector ($D = 49$ in this study) and N is the number of the superpixels.

B. Saliency Changed Area Extraction

The DI is considered to be a combination of the unchanged background and the changed foreground. Thus, the change feature matrix $\mathbf{F}$ can be decomposed into two parts: low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$, where $\mathbf{L}$ corresponds to the background and $\mathbf{S}$ indicates the changed foreground. The basic

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LRMD model is formulated as follows [46]:

$$(\mathbf{L},\mathbf{S}) = \arg\min_{\mathbf{L},\mathbf{S}} \operatorname{rank}(\mathbf{L}) + \lambda\|\mathbf{S}\|_0 \quad \text{s.t. } \mathbf{F} = \mathbf{L} + \mathbf{S} \tag{3}$$

where $\|\cdot\|_0$ is the ℓ_0-norm. Since it is an NP-hard problem, the alternative convex surrogate to solve that is

$$(\mathbf{L},\mathbf{S}) = \arg\min_{\mathbf{L},\mathbf{S}} \|(\mathbf{L})\|_* + \lambda\|\mathbf{S}\|_1 \quad \text{s.t. } \mathbf{F} = \mathbf{L} + \mathbf{S} \tag{4}$$

where $\|\cdot\|_*$ is the nuclear norm and it is a convex relaxation of the matrix rank function. $\|\cdot\|_1$ denotes the ℓ_1-norm to promote sparsity, and the tradeoff parameter $\lambda > 0$ is used to balance the two items. $\mathbf{L}$ and $\mathbf{S}$ can be decomposed precisely using the RPCA [47], [48] algorithm.

After decomposition, we can get the saliency DI from the obtained sparse matrix $\mathbf{S}$. Specifically, the ℓ_1-norm of each column $\mathbf{s_i}$ in $\mathbf{S}$ is used to evaluate the change saliency of the corresponding superpixel. The larger of the $\|\mathbf{s_i}\|_1$, the higher the value assigned to the ith superpixel. A gray-scale change probability map is then generated after normalization.

Although the above model improves robustness to the speckle noise, all the superpixels in saliency DI are independent, and it neglects the spatial relations between neighboring superpixels. It does not ensure that the entire changed region is detected. Meanwhile, it is hard to reduce the coherence between the low-rank part and the sparse part under complex changes. To address this issue, we hope to introduce a constraint to enhance the spatial correlation and enlarge the difference between the two items. The optimized model can be represented as follows:

$$(\mathbf{L},\mathbf{S}) = \arg\min_{\mathbf{L},\mathbf{S}} \|(\mathbf{L})\|_* + \lambda\|\mathbf{S}\|_1 + \gamma\Theta(\mathbf{L}+\mathbf{S}) \quad \text{s.t. } \mathbf{F} = \mathbf{L} + \mathbf{S} \tag{5}$$

where $\Theta(\cdot)$ is the introduced regularization term to make it easier to separate the changed area from the background and γ is a tradeoff parameter.

1) Laplacian Regularization: According to the local invariance assumption [49], if two neighboring superpixels are close to the change features, the representations of these two superpixels in the new subspace should also be close to each other. The spatial correlation constraint can be represented through a similarity matrix $\mathbf{W} = (w_{i,j}) \in \mathbb{R}^{N\times N}$. Specifically, a graph $G(V, E)$ is constructed on the segmented DI, where the node $v_i \in V$ indicates a superpixel and the edge $e_{i,j} \in E$ connects a pair of neighboring superpixels. The edges E are weighted by $\mathbf{W}$, where the weight $w_{i,j}$ of the edge $e_{i,j}$ connecting nodes v_i and v_i is defined as follows:

$$w_{i,j} = \begin{cases} \exp\left(-\dfrac{\|f_i - f_j\|^2}{2^\sigma}\right), & \text{if } (P_i, P_j) \in E \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where P_i and P_j are superpixel patches and σ is a constant that controls the weight. With the above similarity matrix $\mathbf{W}$, the regularization can be defined as follows:

$$\Theta(\mathbf{L},\mathbf{S}) = \frac{1}{2}\sum_{i.j=1}^{N} \|\mathbf{s}_i - \mathbf{s}_j\|_2^2 w_{i,j} = \operatorname{tr}(\mathbf{SDS^T}) - \operatorname{tr}(\mathbf{SWS^T}) = \operatorname{tr}(\mathbf{SMS^T}) \tag{7}$$

where $\operatorname{tr}(\cdot)$ denotes the trace of a matrix, $\mathbf{D} = \operatorname{diag}\{d_{1,1}, \ldots, d_{n,n}\}$ is a degree matrix, and $d_{i,i} = \sum_j w_{i,j}$. $\mathbf{M}$ is a Laplacian matrix, where $\mathbf{M} = \mathbf{D} - \mathbf{W}$.

Indeed, the Laplacian regularization promotes superpixels within the same semantic region (changed and unchanged) to get similar salient values through smoothing the vectors in $\mathbf{S}$. Therefore, it enlarges the feature distance between two subspaces [50].

2) Optimization: Combine Laplacian regularization term with the basic LRMD model, (5) becomes

$$(\mathbf{L},\mathbf{S}) = \arg\min_{\mathbf{L},\mathbf{S}} \|(\mathbf{L})\|_* + \lambda\|\mathbf{S}\|_1 + \gamma\operatorname{tr}(\mathbf{ZMZ^T}) \quad \text{s.t. } \mathbf{F} = \mathbf{L} + \mathbf{S}, \quad \mathbf{S} = \mathbf{Z} \tag{8}$$

where $\mathbf{Z}$ is an auxiliary variable to make the objective function separable.

The optimization problem in (8) can be solved using the ADMM algorithm. Construct the augmented Lagrangian function $\mathcal{L}$ as follows:

$$\begin{aligned}\mathcal{L}(\mathbf{L},\mathbf{S},\mathbf{Z},\mathbf{Y_1},\mathbf{Y_2},\mu) = {} & \|(\mathbf{L})\|_* + \lambda\|\mathbf{S}\|_1 + \gamma\operatorname{tr}(\mathbf{ZMZ^T}) \\ & + \operatorname{tr}(\mathbf{Y_1^T F - L - S}) + \operatorname{tr}(\mathbf{Y_2^T S - Z}) \\ & + \frac{\mu}{2}\left(\|\mathbf{F - L - S}\|_F^2 + \|\mathbf{S - Z}_F^2\|\right)\end{aligned} \tag{9}$$

where $\mathbf{Y_1}$ and $\mathbf{Y_2}$ are the Lagrange multipliers, and $\mu > 0$ is the penalty parameter. The variables $\mathbf{L}, \mathbf{S}, \mathbf{Z}$, $\mathbf{Y_1}$, $\mathbf{Y_2}$, and μ can be optimized iteratively by minimizing the augmented Lagrangian function $\mathcal{L}$, and in each iteration, the variables are updated alternately. This method provides a closed-form solution for each variable and reduces the computational load of the LRMD model.

Updating $\mathbf{L}$*:* Fix $\mathbf{S}$ and $\mathbf{Z}$, $\mathbf{L^{k+1}}$ at the $(k+1)$th iteration can be updated as follows:

$$\begin{aligned}\mathbf{L^{k+1}} &= \arg\min_{\mathbf{L}} \mathcal{L}(\mathbf{L}, \mathbf{S^k}, \mathbf{Z^k}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k) \\ &= \arg\min_{\mathbf{L}}\left(\|\mathbf{L}\|_* + \operatorname{tr}\left(\left(\mathbf{Y_1^k}\right)^{\mathbf{T}}(\mathbf{F - L - S^k})\right) + \frac{\mu^k}{2}\|\mathbf{F - L - S^k}\|_F^2\right) \\ &= \Gamma_{\frac{1}{\mu^k}}\left(\mathbf{F - S^k} + \frac{\mathbf{Y_1^k}}{\mu^{\mathbf{k}}}\right) = \mathbf{U}\Lambda_{\frac{1}{\mu^k}}[\Sigma]\mathbf{V^T}\end{aligned} \tag{10}$$

where $\Gamma_\epsilon(\mathbf{X}) = \mathbf{U}\Lambda_\epsilon[\Sigma]\mathbf{V^T}$, and $\mathbf{U}\Sigma\mathbf{V^T}$ is the singular value decomposition of matrix $\mathbf{X}$. $\Lambda_\epsilon(\cdot)$ is the contraction operator, which is defined as follows:

$$(\Lambda_\epsilon[\mathbf{X}])_{i,j} = \begin{cases} |\mathbf{X}_{i,j}| - \epsilon, & \text{if } |\mathbf{X}_{i,j}| \geq \epsilon \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

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Algorithm 1 ADMM to Optimize the Laplacian Constraint LRMD Model

Input: Change feature matrix $\mathbf{F}$; Parameters λ and γ; Number of superpixels $\mathbf{N}$;

Initialize: $\mathbf{L^0} = \mathbf{S^0} = \mathbf{Z^0} = \mathbf{Y_1^0} = \mathbf{Y_2^0} = 0$; $\mu^0 = 0.1; \mu_{max} = 10^{10}$; $\rho = 1.1$; $\varepsilon_1 = \varepsilon_2 = 10^{-8}$;

1: **while** not converged **do**
2: $\mathbf{L^{k+1}} = \arg\, min_{\mathbf{L}} \mathcal{L}(\mathbf{L}, \mathbf{S^k}, \mathbf{Z^k}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k)$
3: $\mathbf{S^{k+1}} = \arg\, min_{\mathbf{S}} \mathcal{L}(\mathbf{L^{k+1}}, \mathbf{S}, \mathbf{Z^k}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k)$
4: $\mathbf{Z^{k+1}} = \arg\, min_{\mathbf{Z}} \mathcal{L}(\mathbf{L^{k+1}}, \mathbf{S^{k+1}}, \mathbf{Z}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k)$
5: $\mathbf{Y_1^{k+1}} = \mathbf{Y_1^k} + \mu^k(\mathbf{F} - \mathbf{L^{k+1}} - \mathbf{S^{k+1}})$
6: $\mathbf{Y_2^{k+1}} = \mathbf{Y_2^k} + \mu^k(\mathbf{F} - \mathbf{L^{k+1}} - \mathbf{S^{k+1}})$
7: $\mu^{k+1} = \min(\rho\mu^k, \mu_{max})$
8: Check the convergence conditions:
$$\|\mathbf{F} - \mathbf{L_k} - \mathbf{S_k}\|_F / \|\mathbf{F}\|_F < \varepsilon_1$$
$$\max(\|\mathbf{S_k} - \mathbf{S_{k-1}}\|_F, \|\mathbf{L_k} - \mathbf{L_{k-1}}\|_F) < \varepsilon_2$$
9: **end while**

Return: $\mathbf{L_k}$, $\mathbf{S_k}$.

Updating $\mathbf{S}$*:* Fix $\mathbf{L}$ and $\mathbf{Z}$, $\mathbf{S^{k+1}}$ at the $(k+1)$th iteration can be updated as follows:

$$\begin{aligned}\mathbf{S^{k+1}} &= \arg\min_{\mathbf{S}} \mathcal{L}\left(\mathbf{L^{k+1}}, \mathbf{S}, \mathbf{Z^k}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k\right) \\ &= \arg\min_{\mathbf{S}} \Bigg(\lambda\|\mathbf{S}\|_1 + \mathrm{tr}\Big(\left(\mathbf{Y_1^k}\right)^{\mathbf{T}}(\mathbf{F} - \mathbf{L^k} - \mathbf{S})\Big) \\ &\qquad + \mathrm{tr}\Big(\left(\mathbf{Y_2^k}\right)^{\mathbf{T}}(\mathbf{S} - \mathbf{Z^k})\Big) \\ &\qquad + \frac{\mu^k}{2}\Big(\left\|\mathbf{F} - \mathbf{L^{k+1}} - \mathbf{S}\right\|_F^2 + \left\|\mathbf{S} - \mathbf{Z^k}\right\|_F^2\Big)\Bigg) \\ &= \Lambda_{\frac{1}{\mu^k}}\left(\mathbf{F} - \mathbf{L^{k+1}} + \mathbf{Z^k} - \frac{\mathbf{Y_2^k} - \mathbf{Y_1^k}}{\mu^k}\right). \end{aligned} \tag{12}$$

Updating $\mathbf{Z}$*:* Fix $\mathbf{L}$ and $\mathbf{S}$, $\mathbf{Z^{k+1}}$ at the $(k+1)$th iteration can be updated as follows:

$$\begin{aligned}\mathbf{Z^{k+1}} &= \arg\min_{\mathbf{Z}} \mathcal{L}\left(\mathbf{L^{k+1}}, \mathbf{S^{k+1}}, \mathbf{Z}, \mathbf{Y_1^k}, \mathbf{Y_2^k}, \mu^k\right) \\ &= \arg\min_{\mathbf{Z}} \Bigg(\gamma\mathrm{tr}\left(\mathbf{ZMZ^T}\right) + \mathrm{tr}\Big(\left(\mathbf{Y_2^k}\right)^{\mathbf{T}}(\mathbf{S^{k+1}} - \mathbf{Z})\Big) \\ &\qquad + \frac{\mu^k}{2}\left\|\mathbf{S^{k+1}} - \mathbf{Z}\right\|_F^2\Bigg) \\ &= \left(\mu^k\mathbf{S^{k+1}} + \mathbf{Y_2^k}\right)\left(\mu^k\mathbf{I} + 2\gamma\mathbf{M}\right). \end{aligned} \tag{13}$$

Updating $\mathbf{Y_1}$*,* $\mathbf{Y_2}$*, and* μ

$$\mathbf{Y_1^{k+1}} = \mathbf{Y_1^k} + \mu^k\left(\mathbf{F} - \mathbf{L^{k+1}} - \mathbf{S^{k+1}}\right) \tag{14}$$
$$\mathbf{Y_2^{k+1}} = \mathbf{Y_2^k} + \mu^k\left(\mathbf{F} - \mathbf{L^{k+1}} - \mathbf{S^{k+1}}\right) \tag{15}$$
$$\mu^{k+1} = \min(\rho\mu^k, \mu_{\max}). \tag{16}$$

The stop criteria at the kth iteration can be expressed as follows:

$$\|\mathbf{F} - \mathbf{L_k} - \mathbf{S_k}\|_F / \|\mathbf{F}\|_F < \varepsilon_1 \tag{17}$$
$$\max(\|\mathbf{S_k} - \mathbf{S_{k-1}}\|_F, \|\mathbf{L_k} - \mathbf{L_{k-1}}\|_F) < \varepsilon_2. \tag{18}$$

The main steps of solving the optimization problem in (8) is summarized in Algorithm 1, and the detailed derivation can be seen in Appendix A.

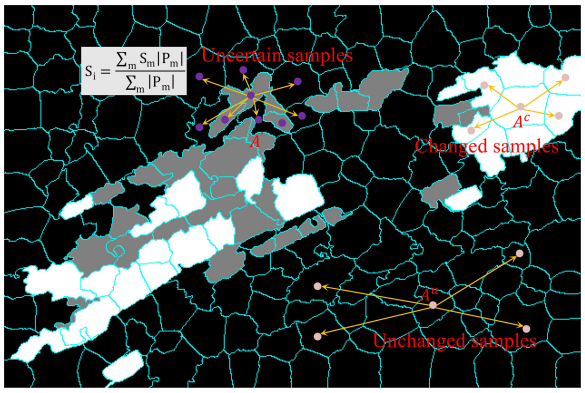


Fig. 2. Illustration of the local spatial correlations in saliency DI. All the positive samples (in white), negative samples (in black), and uncertain samples (in gray) are selected from the saliency DI and the yellow lines connect those superpixels that have local spatial correlations.

C. Classification Refinement

As mentioned above, the Laplacian constraint LRMD model enhanced the separability between changed and unchanged superpixels. There are still masses of fuzzy superpixels that are hard to distinguish accurately. To address this problem and further improve the detection accuracy, those inexact superpixels are refined by learning the projection from the original change feature matrix to saliency DI [51]. The refining process is considered an unsupervised joint optimization problem for both certain and uncertain samples. In the following, we will describe the constructed CR module in detail.

After obtaining the sparse matrix S, the changed and unchanged regions can be roughly distinguished. Thus, high-confident superpixels can be selected based on simple, but effective threshold segmentation. We set two thresholds, τ_1 and τ_2 ($\tau_1 < \tau_2$), to classify superpixels into three groups: the changed class as the positive samples, the unchanged class as the negative samples, and the intermediate class as the uncertain samples. Then, the high-confident sample matrix $\mathbf{A} \in \mathbb{R}^{N_1 \times D}$ is constructed, which contains both positive and negative samples, where N_1 is the number of high-confident samples. $\mathbf{B} = [b_1, b_2, \ldots, b_{N_1}] \in \mathbb{R}^{N_1 \times 2}$ is the corresponding label matrix. For positive samples, the label vector b is set to $[1, 0]$; for negative samples, b is set to $[0, 1]$.

For those uncertain superpixels, the local spatial correlations are utilized to determine their salient value, as shown in Fig. 2, which is calculated by the following equation:

$$\tilde{\mathbf{s}}_\mathbf{i} = \frac{\sum_m \mathbf{s_m}|P_m|}{\sum_m |P_m|} \tag{19}$$

where m is the number of superpixels adjacent to the ith uncertain sample, and $|P_m|$ is the number of pixels within this sample. Similarly, the uncertain sample matrix $\mathbf{A}' \in \mathbb{R}^{N_2 \times D}$ and $\mathbf{B}' = [b'_1, b'_2, \ldots, b'_{N_2}] \in \mathbb{R}^{N_2 \times 2}$ can be constructed, and the label vector is formulated as $b'_i = [\tilde{\mathbf{s}}_\mathbf{i}, 1 - \tilde{\mathbf{s}}_\mathbf{i}]$.

Both the high-confident and uncertain samples are used in the unsupervised projection learning, and then the optimized

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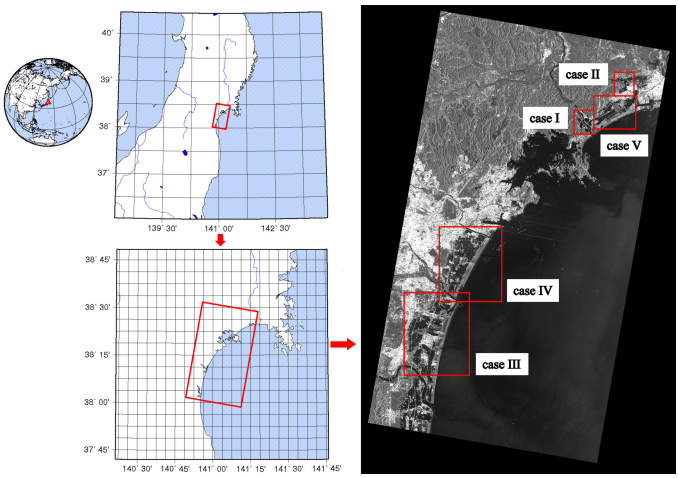


Fig. 3. Geographic location of the case area.

model is built as follows:

$$\mathbf{P} = \arg\min_{\mathbf{P}} \left\{ \frac{1}{2}\|\mathbf{P}\|_F^2 + \frac{\theta_1}{2}\|\mathbf{AP} - \mathbf{B}\|_F^2 + \frac{\theta_2}{2}\left\|\mathbf{A}'\mathbf{P} - \mathbf{B}'\right\|_F^2 \right\} \tag{20}$$

where $\mathbf{P} \in \mathbb{R}^{D\times 2}$ is the projection matrix to be learned and θ_1 and θ_2 are balancing parameters. It should be noted that the background area is usually much larger than the changed area, which leads to a problem of class imbalance. To deal with this issue, sample weight is added to balance the contributions of the different class samples in projection learning. Equation (20) can be updated as follows:

$$\mathbf{P} = \arg\min_{\mathbf{P}} \left\{ \frac{1}{2}\|\mathbf{P}\|_F^2 + \frac{\theta}{2}\sum_{i=1}^{N_1+N_2} \tilde{w}_i \left\|\tilde{\mathbf{A}}_\mathbf{i}\mathbf{P} - \tilde{\mathbf{B}}_\mathbf{i}\right\|_F^2 \right\} \tag{21}$$

where $\tilde{w}_i$ is the weight for the ith sample (positive, negative, or uncertain). Considering that there are higher numbers of negative samples than positive samples, the positive ones are assigned the larger weights, while the negative ones have the lower weights. Specifically, $\tilde{w}_i$ is set to be

$$\tilde{w}_i = \begin{cases} 0.5, & \text{Uncertain samples} \\ 1, & \text{Negative samples} \\ \dfrac{N_n}{N_p}, & \text{Positive samples} \end{cases} \tag{22}$$

where N_n and N_p are the numbers of the negative and positive samples, respectively. The closed-form solution of (21) is

$$\mathbf{P} = \left(\mathbf{I} + \theta\tilde{\mathbf{A}}^\mathbf{T}\tilde{\mathbf{W}}\tilde{\mathbf{A}}\right)^{-1}\left(\lambda\tilde{\mathbf{A}}^\mathbf{T}\tilde{\mathbf{W}}\tilde{\mathbf{B}}\right). \tag{23}$$

$\tilde{\mathbf{W}}$ is a weight matrix where $\tilde{\mathbf{W}}_{\mathbf{i},\mathbf{i}} = \tilde{w}_i$ and $\mathbf{I}$ is an identity matrix. After learning the projection, the refined salient values of the uncertain samples can be calculated by $\mathbf{A}'\mathbf{P}$.

IV. Experiment

To fully evaluate the performance of the proposed method, a series of experiments are conducted on five real-world datasets. We first compare the proposed method with other advanced CD methods to verify its superiority. Then, the effectiveness of the architecture is demonstrated through the ablation study. Discussions about the parameter sensitivity and computational complexity are shown at the end.

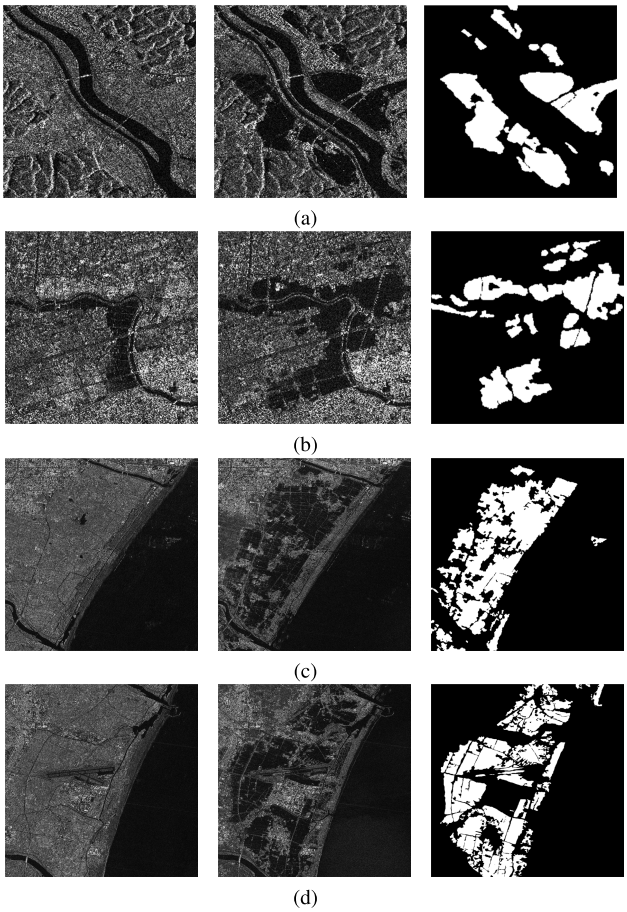


Fig. 4. Bititemporal images and corresponding reference images of the Sendai datasets. (a) Dataset I. (b) Dataset II. (c) Dataset III. (d) Dataset IV.

A. Dataset and Experimental Setting

The multitemporal data used in this study are from the TerraSAR-X satellite sensor, which has a spatial resolution of 9 × 9 m after preprocessing in the stripmap mode. The case area mainly belongs to Sendai, Japan, as shown in Fig. 3. On March 11, 2011, a strong earthquake and subsequent tsunami attacked the coastal areas of Sendai. The tsunami caused devastating damage, and a large swath of land was drowned [18]. The used data consists of four images, acquired on September 21, 2008, October 20, 2010, March 13, 2011, and March 23, 2011, respectively. Five datasets are made from those coastal areas, which can evaluate the effectiveness and robustness of the proposed framework under multiscale and multitemporal conditions. Fig. 4 shows four bitemproal image pairs and Fig. 5 shows the short time-series dataset. As can be seen in Fig. 4, Datasets I and II are two small-scale datasets, and their sizes are 286 × 298 and 308 × 280, respectively. Datasets III and IV are two large-scale datasets, and their sizes are 1001 × 1001 and 941 × 901, respectively. Dataset V contains four historical images, and the size is 635 × 422. It should be noted that all the production of the ground-truth maps references Google historical images, and the reference change regions are manually marked pixel by pixel.

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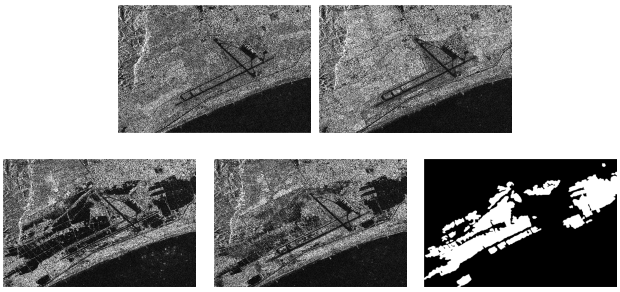


Fig. 5. Multititemporal images and corresponding reference image of Sendai Dataset V.

To assess the performance of the proposed method, some quantitative evaluation indices are adopted, which are derived from the confusion matrix, including false negatives (FNs), false positives (FPs), the overall error (OE), the overall accuracy (OA), F1 score (F1), and Kappa coefficient (KC). For our method, we set the number of superpixels $N = 450$, initialization for variables and parameters in the improved LRMD module are set as $\mathbf{L^0} = \mathbf{S^0} = \mathbf{Z^0} = \mathbf{Y_1^0} = \mathbf{Y_2^0} = 0$, $\mu^0 = 0.1$, $\mu_{\max} = 1e10$, and $\rho = 1.1$. Regularization parameters in the model are set as $\lambda = 0.3$, $\gamma = 1.0$, and $\theta = 10$. All the parameter values are kept constant throughout the experiments except for parametric analysis. Detailed discussions about the parameter sensitivity are provided in Section IV-D. All the experiments were conducted using MATLAB 2020b on an Intel i5-10400 CPU with 32 GB RAM.

B. Comparison With State-of-the-Arts

To verify the superiority of the proposed model, a thorough comparison with recently proposed CD methods was made. These comparison methods include.

1) *LR Operator:* As a classical SAR PBCD method, the LR operator has been widely used in CD tasks. In this study, we treat it as a baseline method for performance comparison. Note that the K-means clustering algorithm is implemented on the DI after conducting the LR operator to generate the final binary change map.
2) *PCANet [52]:* A preclassification strategy is designed to obtain the high-confident labeled pixels, and the image patches centered at these pixels are then generated to train the PCANet model.
3) *CWNN [53]:* A dual-tree complex wavelet transform is introduced into convolutional neural networks (CNNs) for changed and unchanged pixel classification. A virtual sample generation strategy is employed to create samples for CWNN training.
4) *NSCT-NLM [18]:* The nonsubsampled contourlet transform (NSCT) image fusion algorithm is employed in the original DI images, which are generated from the LR operator and the slow feature analysis (SFA) algorithm. The nonlocal means (NLMs) filtering method is then adopted in the fused DI to suppress the speckle noise.
5) *MSDC-SAM [54]:* K-means++ clustering is combined with CNNs to construct an unsupervised end-to-end difference representation learning framework. The self-attention module (SAM) is introduced to mine important spatial structure features.
6) *RUSACD [39]:* A multiscale superpixel reconstruction method is developed to generate DI, and a generative adversarial network is adopted to increase the pseudo-label samples. CWNN is trained to classify the fuzzy samples.

Except for the LR and NSCT-NLM algorithms, the other four comparisons are all deep-learning-based. Essentially, the above comparison algorithms are all unsupervised. These deep-learning-based ones are implemented by using the source codes provided by the authors with default parameters. Note that the first five are PBCD methods and the last one is an OBCD method. Since these introduced methods are bitemporal-based, we use the first four bitemporal datasets to conduct comparison experiments.

Fig. 6 shows the classification results using different CD methods. In binary change maps, the red and cyan areas denote the false alarm and missed detection, respectively. Visually, the proposed method performed better than others, no matter for small-scale or large-scale datesets. There are fewer holes and gaps inside the changed regions, which means that the changed components extracted are more complete. Meanwhile, the less isolated changed patches mean a lower false alarm. On the contrary, a lot of isolated noise points remained in the binary change maps generated by LR, CWNN, and MSDC-SAM, for example, Dataset I for LR, Dataset IV for CWNN, and Dataset III for MSDC-SAM. While for PCANet, NSCT-NLM, and RUSACD, there are some large-area missed detections, such as Dataset I for PCANet, Dataset II for NSCT-NLM, and Dataset IV for RUSCAD. Fig. 7 presents the statistics of FN, FP, and OE using different methods in the form of histograms. We can see that the proposed method outperforms other methods on all four datasets as it keeps the lowest values of OE. From the perspective of numerical analysis, it can also restrain the FN and FP at a lower level, which is consistent with the above visualization analysis. On the whole, it presents a better compromise between false alarms and missed detections.

The quantitative comparison results for OA, F1, and KC are summarized in Table I, where the two optimal values of each evaluation index are shown in bold. As is listed in Table I, the values of three evaluation metrics for our method rise by a large margin compared with other methods. Especially for Datasets II and IV, the KC increases by 3.46% and 3.63%, respectively. This is mainly due to the strong capabilities in suppressing the speckle noise and extracting the more complete changed regions for our method. The introduction of Laplacian regularization reduces the missed detection of those slight changes, and the refined module further lowers the probability of misclassification for uncertain superpixels.

To further verify the robustness of the proposed method, an additional experiment was conducted on Dataset V. Fig. 8 shows the classification result, and the evaluation results are listed in Table II. It can be seen that our method also performs well on multitemporal data, which demonstrates its excellent ability to monitor constant changes.

C. Ablation Study

Based on the LRMD model, the proposed framework integrates a Laplacian regularization and a CR module. In this

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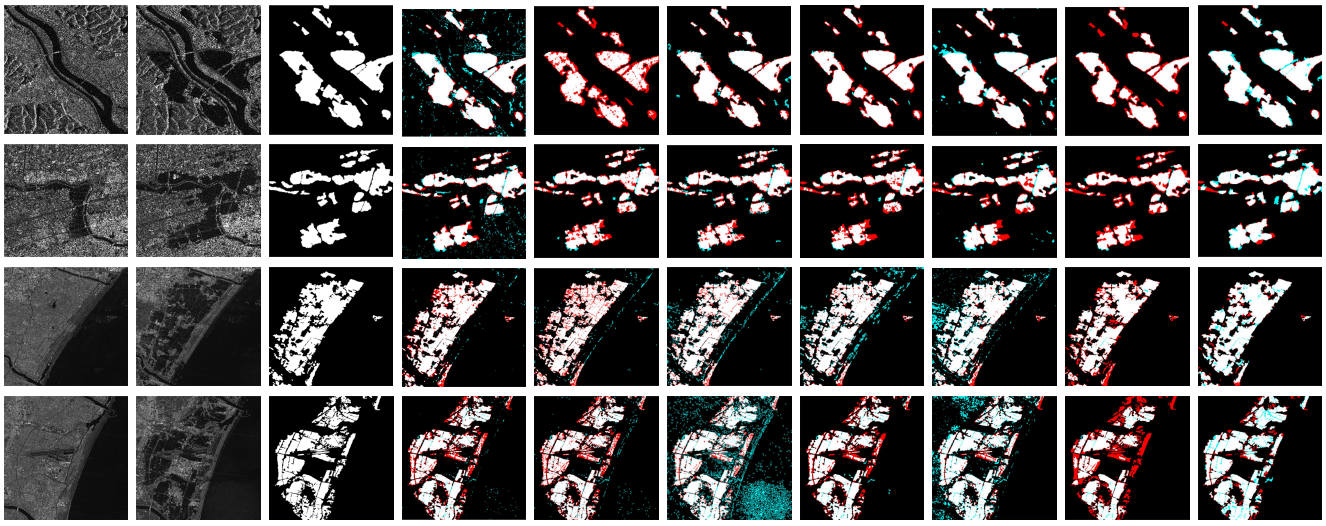


Fig. 6. Visual results of different models on Sendai datasets. From top to bottom, they correspond to Datasets I–IV. White represents the true positive (TP), black is the true negative (TN), cyan indicates the FP, and red stands as the FN.

TABLE I
RESULTS OF QUANTITATIVE EVALUATION ON THE SENDAI DATASETS

Methods	Dataset I			Dataset II			Dataset III			Dataset IV		
	OA ↑	F1↑	KC↑	OA↑	F1↑	KC↑	OA↑	F1↑	KC↑	OA↑	F1↑	KC↑
LR [16]	0.9417	0.8689	0.8316	**0.9442**	**0.8537**	**0.8192**	**0.9602**	**0.9037**	**0.8788**	0.9331	0.8414	0.7996
PCANet [50]	0.9221	0.7740	0.7298	0.9430	0.8353	0.8014	0.9465	0.8693	0.8359	0.9307	0.8372	0.7937
CWNN [51]	**0.9646**	**0.9122**	**0.8901**	0.9442	0.8444	0.8107	0.9454	0.8779	0.8427	0.8402	0.7174	0.6107
NSCT-NLM [18]	0.9608	0.8989	0.8748	0.9329	0.7974	0.7586	0.9468	0.8830	0.8487	**0.9348**	**0.8454**	**0.8047**
MSDC-SAM [52]	0.9601	0.9044	0.8792	0.9376	0.8229	0.7855	0.9407	0.8729	0.8344	0.9163	0.8332	0.7777
RUSACD [29]	0.9486	0.8620	0.8311	0.9284	0.7772	0.7367	0.9509	0.8771	0.8468	0.9025	0.7429	0.6867
Proposed	**0.9694**	**0.9286**	**0.9091**	**0.9546**	**0.8820**	**0.8538**	**0.9694**	**0.9304**	**0.9108**	**0.9416**	**0.8739**	**0.8359**

section, ablation experiments are carried out on all datasets to verify the validity of both modules.

To clarify the effects of the Laplacian regularizer, we first compute the normalization distance between the feature subspace of the sparse term $\mathbf{S}$ and the low-rank term $\mathbf{L}$. Fig. 9 reports the changes in subspace distance before and after using Laplacian regularization on five datasets. As expected, the subspace distance is enlarged significantly after integrated Laplacian regularization, especially on Datasets II and V. It verifies that the Laplacian regularization boosts the difference between the changed regions and the background.

To further understand the functions of two components in the proposed framework, the generated DIs in different steps are visualized in Fig. 10. Intuitively, the original LR-based DI has lots of noise in the background, and the separability between the changed areas and the background is not particularly good. The saliency DIs generated by the proposed improved LRMD, by contrast, present more clean backgrounds and more completed change areas. The saliency of uncorrelated noise in the background is suppressed significantly by extracting the sparse term. Meanwhile, the application of Laplacian regularization reduces the difference between the locally similar superpixels and enhances the differences between those heterogeneous superpixels. The refinement module further helps to improve the separation, especially for those uncertain regions. After refinement, the binary change maps can be generated with threshold segmentation or some clustering methods.

TABLE II
RESULTS OF QUANTITATIVE EVALUATION ON DATASET V

Dataset	Method	OA↑	F1↑	KC↑
Dataset V	Proposed	0.9636	0.9073	0.8847

Table III lists the quantitative evaluation results of the original LRMD (we use RPCA to optimize this model) and its variants. It can be seen that the three metrics in all scenes for the baseline method are the lowest. Introducing Laplacian (LP) regularization significantly increases the performance compared with the baseline. In particular, model I improves Kappa values by 31% and 33% on Datasets I and III, respectively. It fully demonstrated that Laplacian regularization is effective in our proposed framework. The other interesting finding is that model I performs much better than model II, which indicates that the effect is less distinct when directly applying the CR module to the baseline without Laplacian regularization.

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TABLE III
ABLATION STUDY OF THE PROPOSED FRAMEWORK

Metrics	RPCA	LP.	CR.	Dataset I	Dataset II	Dataset III	Dataset IV	Dataset V
OA↑	√			0.8369	0.8570	0.8264	0.8431	0.8718
	√	√		0.9660	0.9508	0.9630	0.9371	0.9551
	√		√	0.8978	0.9511	0.9397	0.8537	0.9541
	√	√	√	**0.9694**	**0.9546**	**0.9694**	**0.9416**	**0.9636**
F1↑	√			0.6919	0.7017	0.6716	0.7141	0.7341
	√	√		0.9199	0.8700	0.9159	0.8660	0.8817
	√		√	0.7964	0.8741	0.8714	0.7477	0.8918
	√	√	√	**0.9286**	**0.8820**	**0.9304**	**0.8739**	**0.9073**
KC↑	√			0.5870	0.6116	0.5579	0.6092	0.6514
	√	√		0.8983	0.8397	0.8922	0.8249	0.8544
	√		√	0.7304	0.8437	0.8322	0.6500	0.8626
	√	√	√	**0.9091**	**0.8538**	**0.9108**	**0.8359**	**0.8847**

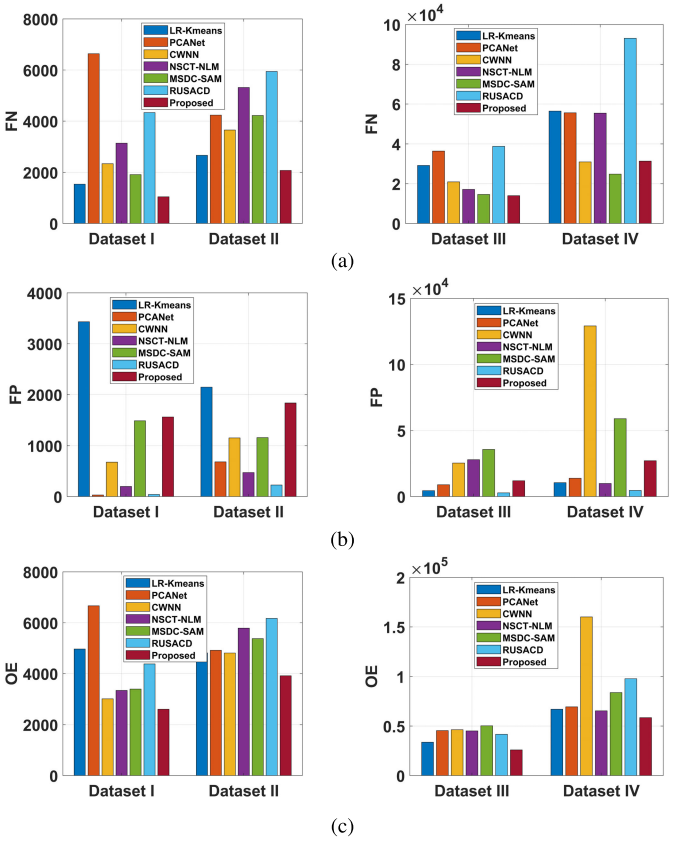


Fig. 7. Comparison of three quantitative evaluation indices on four datasets. (a) Values of FNs for different methods. (b) Values of FPs for different methods. (c) Values of OE for different methods.

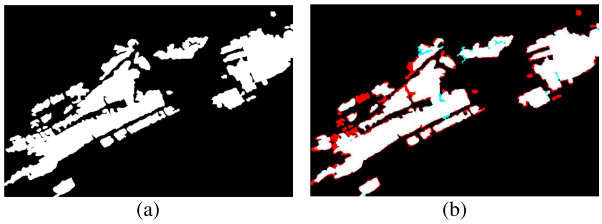


Fig. 8. Classification results of Dataset V using the proposed method. (a) Ground truth. (b) Binary change map.

Compared with model I, the addition of the refinement module further improves the classification accuracy. For example, the KC improves by 1.86% and 3.03% on Datasets III and V, respectively. The results indicate that the refinement module can enhance the discrimination and optimize the classification results of the model.

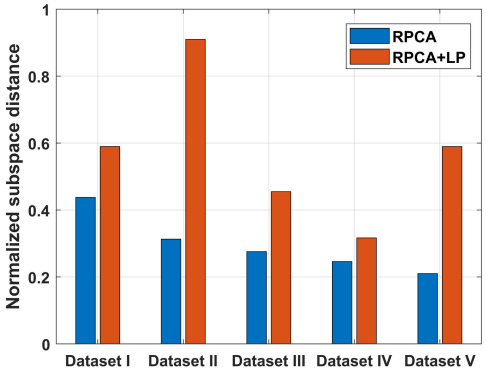


Fig. 9. Comparison of subspace distribution estimated on five datasets before and after using Laplacian regularization.

D. Parameter Analysis

There are two main classes of parameters in the proposed method: the parameters used in LRMD and the refining process. The former mainly includes the number of superpixels N and the tradeoff parameters λ and γ. The latter mainly includes the balancing parameter θ and the threshold parameters τ_1 and τ_2. To evaluate their influence on classification accuracy, we analyze the relationships between KCs and their values on five datasets. Experiments are conducted by tuning one parameter while fixing another.

1) Parameter N: Parameter N is usually selected in light of the image scales. Theoretically, the larger N means a smaller segmented granularity, thus the classification is more meticulous. However, it will be very time-consuming if the size is set too large. Fig. 11 shows the curves of accuracy on five datasets as N increases from 200 to 750. We can see that the Kappa values first rise slightly and then gradually decrease. As a whole, these trends are not obvious in small-scale datasets, especially for Datasets I and II. The curves slope downward significantly for Datasets III and IV when N is larger than 600. This is mainly due to the complexity and heterogeneity of the

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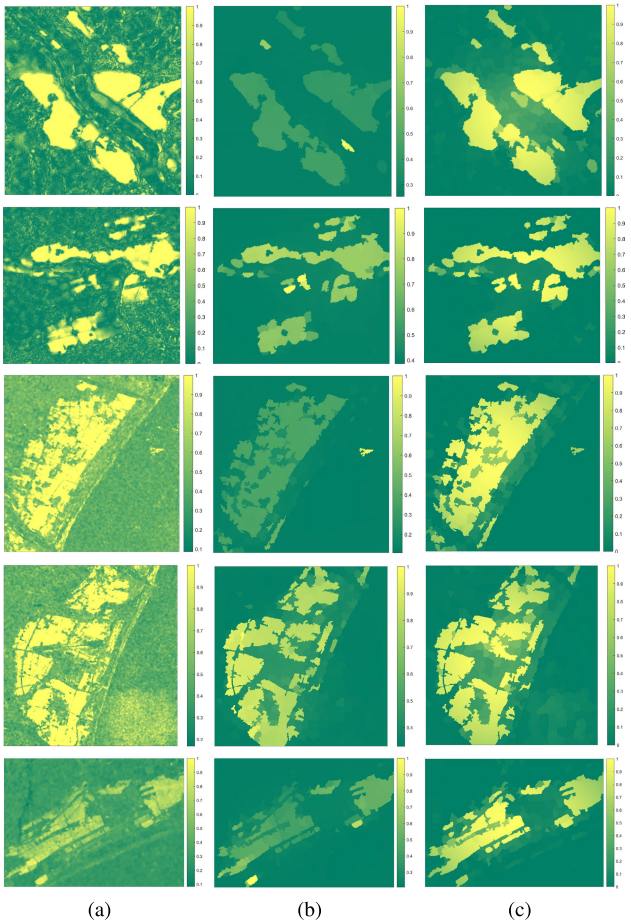


Fig. 10. Visual results of the DIs in different steps on Sendai datasets. From top to bottom, they correspond to Datasets I–V. (a) Original LR-based DIs. (b) DIs generated by Algorithm 1. (c) DIs generated by Algorithm 2.

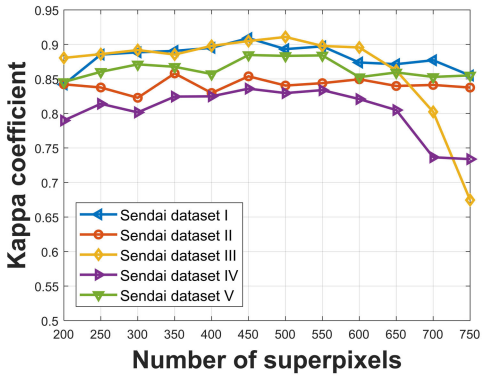


Fig. 11. Classification accuracy on Sendai datasets with different values of the parameter N.

changed scenes. The uncorrelated changes in the background may make a greater difference in the classification decision under the smaller segmented granularity. The larger N not only fails to promote classification, but also brings more errors.

Fig. 12 presents the DIs generated by superpixel segmentation on Datasets I and II with $N = 100, 200, 400$, and 800. It is noticeable that when N is larger, the scales of superpixel patches are smaller, and change details are better preserved. On the contrary, the smaller N is, the larger the size of superpixels, and the block effect is more obvious. To balance the classification accuracy and computing complexity, we selected a compromise value of $N = 450$ as the baseline in our study.

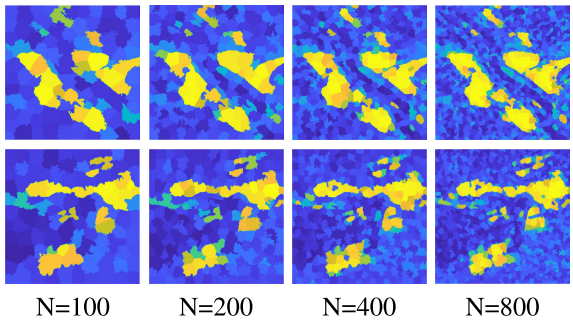


Fig. 12. Comparison of the DIs on Datasets I and II with different values of the parameter N.

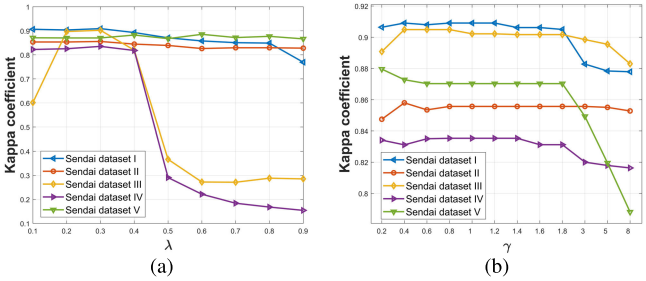


Fig. 13. Sensitivity analysis of the parameters λ and γ.

2) Parameters λ and γ: Parameters λ and γ are used to balance the sparse term and the Laplacian regularization term in the improved LRMD model. In Fig. 13(a), we vary the values of λ from 0.1 to 0.9 with step 0.1 and fix the other parameters. It can be seen that the trends in Kappa values are quite different between the small-scale and large-scale datasets. Specifically, the performance had remained stable as λ increases on small-scale datasets, while the performance deteriorated dramatically as $\lambda > 0.4$ on Datasets III and IV. The reason is that when setting a large λ, the sparse term will obtain more uncorrelated information in the LRMD process. The information includes pseudo-changes and the speckle noise, which severely degrade the refinement and binary classification. Therefore, it is more sensitive to λ for large-scale datasets. To get the best performance on all datasets, we set λ to be 0.3 in this study.

In Fig. 13(b), the values of γ change from 0.1 to 1.8 with step 0.2, and we add three points at 3, 5, and 8. We can find that the performance of our model is generally stable over the range of γ from 0.1 to 1.8 and decreases only when it changes from 1.8 to 8. It demonstrated that the model can work well in a large range of γ settings (we set 1.0 in our study). The Laplacian regularization has strong robustness in multiscale datasets.

3) Parameters θ: Parameters θ are used to balance the regularization term in the refinement module. We can see from Fig. 14 that the curves first stay stable, then increase dramatically when θ increases from 1e-4 to 1e-3, and finally remain stable again. The model performs well over a wide

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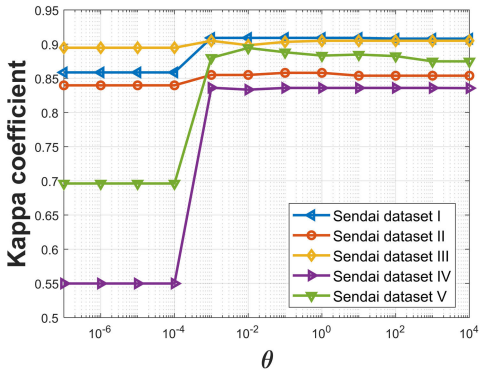


Fig. 14. Sensitivity analysis of the parameter θ.

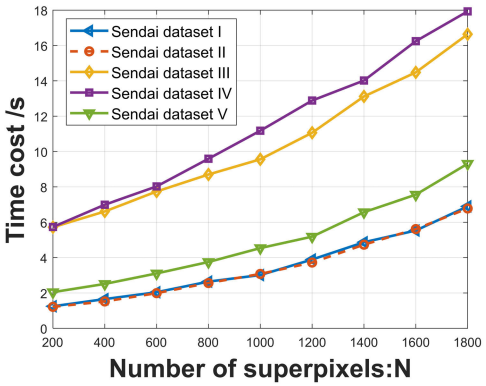


Fig. 16. Influence of the parameter N on time cost.

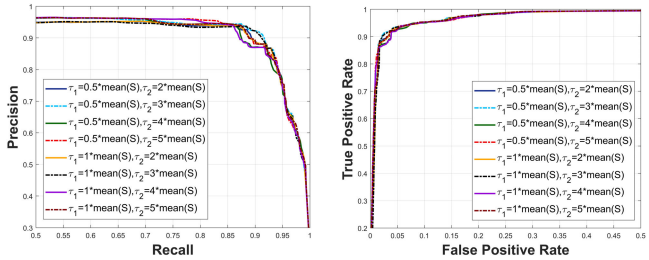


Fig. 15. Evaluation of performance using different thresholding strategies with respect to PR and ROC metrics on Dataset III. (a) PR curve. (b) ROC curve.

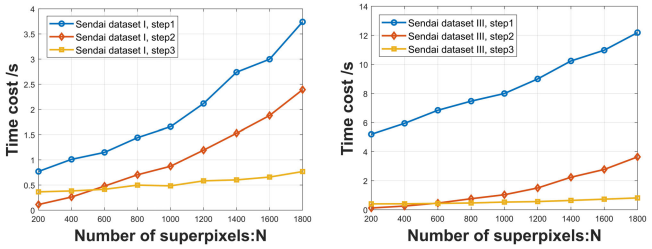


Fig. 17. Comparison of the time costs for different processes of the proposed framework on: (a) Sendai dataset I. (b) Sendai dataset III.

range of θ from 1e-3 to 1e-4, so we set a simple value of $\theta = 10$ in our study.

4) Parameters τ_1 and τ_2: In addition, the sensitivity to the different values of the sample screening thresholds τ_1 and τ_2 in the refining process is examined. Fig. 15(a) and (b) shows the PR curves and ROC curves on Dataset III using different τ_1 and τ_2. We set τ_1 as the 0.5 and 1 times average salient values of DI, and τ_2 is set three–five times, respectively. We can observe that the curves approximated coincide for all permutations, which demonstrates that our method is robust for sample selection.

E. Complexity Analysis

The main computational process of the proposed method consists of three steps: the first step includes superpixel segmentation using SLIC and the construction of the change feature matrix; the second step is optimizing the LRMD model integrated with the Laplacian regularizing constraints; and the third step is solving the joint optimizing function in classification refinement. We calculated the time complexity for the second and third steps in detail and analyzed the effect of the parameter N on the time cost.

For the second step, the time consumption depends mainly on the optimization of Algorithm 1. Here, the used matrices include $\mathbf{L}, \mathbf{S}, \mathbf{Z}, \mathbf{Y_1}, \mathbf{Y_2} \in \mathbb{R}^{D\times N}$. In particular, updating $\mathbf{L}$ needs $\mathcal{O}(DN^2 + D^2N)$, which involves SVD and matrix subtraction. Updating $\mathbf{S}$ requires $\mathcal{O}(DN)$ and updating $\mathbf{Z}$ requires $\mathcal{O}(2DN^2 + D^2N + N^3 + DN) = \mathcal{O}(N^3)$ $(D \ll N)$, which involves matrix subtraction, multiplication, and inversion. Combining three parts, the final time complexity is $\mathcal{O}(N^3)$. For the third step, the time consumption depends mainly on the optimization for (19), which involves matrix addition, multiplication, and inversion. It needs $\mathcal{O}(DN^2 + 3D^2 + D^3 + 2DN) = \mathcal{O}(DN^2)$.

Fig. 16 shows the change in the time cost of the proposed method over the five datasets as the parameter N increases from 200 to 1800. It can be observed that the time cost presents an approximately linear increase with the increase of the parameter N. Meanwhile, they are more time-consuming for two large-scale datasets than the small-scale ones with the same value of N. Fig. 17(a) and (b) plots the time–cost change curves in different steps on Datasets I and III, respectively. It is obvious that the first step is more time-consuming than the second and third steps. This is mainly due to the complexity of the change feature extraction. The other important finding is that the third step has a significantly lower running time than the other steps, and it hardly increases with increasing N, while the time cost of the second step increases significantly. This is consistent with previous time complexity analyses. It demonstrates that the first step is the most time-consuming process within the proposed framework.

V. Conclusion

In this article, an unsupervised superpixel analysis framework based on LRMD is presented for a high-resolution SAR image CD problem. The Laplacian regularizing constrained LRMD model is proposed to generate the saliency DI, which indicates the change probabilities of the segmented superpixels. The introduction of Laplacian regularization enhanced local spatial correlations and improved the separation between changed regions and the background. Meanwhile, an unsupervised learning module is trained to refine the coarse

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saliency DI. Extensive experiments on five real-world datasets demonstrated that our method can achieve encouraging performance compared to the recently developed state-of-the-art methods with both effectiveness and superiority.

For future work, we will consider combining the proposed framework with deep learning-based methods, such as unsupervised contrastive learning. Since high-confidence pseudo-training samples can easily be obtained from the LRMD model, they may assist in the training and optimization processes of the networks. In addition, to further expand the applications of the proposed framework, using multitemporal SAR data with longer time series merits further study.

APPENDIX A
OPTIMIZATION FOR ALGORITHM 1

In this appendix, the detailed update scheme required for optimizing Algorithm 1 is described. First, the augmented Lagrangian function in (9) is rewritten as the minimization of

$$\begin{aligned}\mathcal{L}(\mathbf{L},\mathbf{S},\mathbf{Z},\mathbf{Y_1},\mathbf{Y_2},\mu) &= \|(\mathbf{L})\|_* + \lambda\|\mathbf{S}\|_1 + \gamma\mathrm{tr}\big(\mathbf{ZMZ^T}\big)\\ &+ \frac{\mu}{2}\left\|\mathbf{F}-\mathbf{L}-\mathbf{S}+\frac{\mathbf{Y_1}}{\mu}\right\|_F^2 - \frac{1}{2\mu}\|Y_1\|_F^2\\ &+ \frac{\mu}{2}\left\|\mathbf{S}-\mathbf{Z}+\frac{\mathbf{Y_1}}{\mu}\right\|_F^2 - \frac{1}{2\mu}\|Y_2\|_F^2.\end{aligned} \tag{24}$$

Then, the variables $\mathbf{L},\mathbf{S},\mathbf{Z},\mathbf{Y_1}$ and $\mathbf{Y_2}$ can be updated in turn. Before updating, we introduce two operators.

Operator 1: Assume the variable matrix $\mathbf{X}\in\mathbb{R}^{m\times n}$, for any other matrix $\mathbf{Z}\in\mathbb{R}^{m\times n}$, the following optimization:

$$\min_{\mathbf{X}} \epsilon\|\mathbf{X}\|_1 + \frac{1}{2}\|\mathbf{X}-\mathbf{Z}\|_F^2 \tag{25}$$

has the solution: $\mathbf{X}^* = \Lambda_\epsilon(\mathbf{Z})$, where ϵ is a balancing parameter. $\Lambda_\epsilon(\cdot)$ is the contraction operator, which is defined as follows:

$$(\Lambda_\epsilon[\mathbf{Z}])_{i,j} = \begin{cases} |\mathbf{Z}_{i,j}| - \epsilon, & \text{if } |\mathbf{Z}_{i,j}| \geq \epsilon \\ 0, & \text{otherwise.}\end{cases} \tag{26}$$

Operator 2: Assume the variable matrix $\mathbf{X}\in\mathbb{R}^{m\times n}$, for any other matrix $\mathbf{Z}\in\mathbb{R}^{m\times n}$, the following optimization:

$$\min_{\mathbf{X}} \epsilon\|\mathbf{X}\|_* + \frac{1}{2}\|\mathbf{X}-\mathbf{Z}\|_F^2 \tag{27}$$

has the solution: $\mathbf{X}^* = \Gamma_\epsilon(\mathbf{Z})$, where $\Gamma_\epsilon(\mathbf{Z}) = \mathbf{U}\Lambda_\epsilon[\Sigma]\mathbf{V^T}$, $\Lambda_\epsilon(\cdot)$ is as definition in operator 1.

1) *Update* $\mathbf{L}$

$$\begin{aligned}\mathbf{L^{k+1}} &= \arg\min_{\mathbf{L}} \mathcal{L}\big(\mathbf{L},\mathbf{S^k},\mathbf{Z^k},\mathbf{Y_1^k},\mathbf{Y_2^k},\mu^k\big)\\ &= \arg\min_{\mathbf{L}}\left(\|\mathbf{L}\|_* + \frac{\mu^k}{2}\left\|\mathbf{L}-\left(\mathbf{F}-\mathbf{S^k}+\frac{\mathbf{Y_1^k}}{\mu^k}\right)\right\|_F^2\right)\\ &= \Gamma_{\frac{1}{\mu^k}}\left(\mathbf{F}-\mathbf{S^k}+\frac{\mathbf{Y_1^k}}{\mu^k}\right) = \mathbf{U}\Lambda_{\frac{1}{\mu^k}}[\Sigma]\mathbf{V^T}.\end{aligned} \tag{28}$$

2) *Update* $\mathbf{S}$

$$\begin{aligned}\mathbf{S^{k+1}} &= \arg\min_{\mathbf{S}} \mathcal{L}\big(\mathbf{L^{k+1}},\mathbf{S},\mathbf{Z^k},\mathbf{Y_1^k},\mathbf{Y_2^k},\mu^k\big)\\ &= \arg\min_{\mathbf{S}}\left(\lambda\|\mathbf{L}\|_1 + \frac{\mu^k}{2}\left\|\mathbf{F}-\mathbf{L^{k+1}}-\mathbf{S^k}+\frac{\mathbf{Y_1^k}}{\mu^k}\right\|_F^2 + \frac{\mu^k}{2}\left\|\mathbf{S}-\mathbf{Z^k}+\frac{\mathbf{Y_2^k}}{\mu^k}\right\|_F^2\right)\\ &= \frac{1}{2}\left\|\mathbf{S}-\left(\mathbf{F}-\mathbf{L^{k+1}}+\mathbf{Z^k}-\frac{\mathbf{Y_2^k}-\mathbf{Y_1^k}}{\mu^k}\right)\right\|_F^2 + \frac{\lambda}{2\mu^k}\|\mathbf{S}\|_1\\ &= \Lambda_{\frac{1}{\mu^k}}\left(\mathbf{F}-\mathbf{L^{k+1}}+\mathbf{Z^k}-\frac{\mathbf{Y_2^k}-\mathbf{Y_1^k}}{\mu^k}\right).\end{aligned} \tag{29}$$

3) *Update* $\mathbf{Z}$

$$\begin{aligned}\mathbf{Z^{k+1}} &= \arg\min_{\mathbf{Z}} \mathcal{L}\big(\mathbf{L^{k+1}},\mathbf{S^{k+1}},\mathbf{Z},\mathbf{Y_1^k},\mathbf{Y_2^k},\mu^k\big)\\ &= \arg\min_{\mathbf{Z}}\left(\gamma\mathrm{tr}\big(\mathbf{ZMZ^T}\big) + \frac{\mu^k}{2}\left\|\mathbf{S^{k+1}}-\mathbf{Z}+\frac{\mathbf{Y_2^k}}{\mu^k}\right\|_F^2\right).\end{aligned} \tag{30}$$

Take the partial derivative for two terms in the above equation, respectively

$$\gamma\frac{\partial\mathrm{tr}\big(\mathbf{ZMZ^T}\big)}{\partial\mathbf{Z}} = 2\gamma\mathbf{ZM} \tag{31}$$

$$\frac{\partial\left(\frac{\mu^k}{2}\left\|\mathbf{S^{k+1}}-\mathbf{Z}+\frac{\mathbf{Y_2^k}}{\mu^k}\right\|_F^2\right)}{\partial\mathbf{Z}} = \mu(\mathbf{S}-\mathbf{Z})-\mathbf{Y_2^k}. \tag{32}$$

Let the above equations equal to zero, then we have the following closed-form solution:

$$\mathbf{Z^{k+1}} = \big(\mu^k\mathbf{S^{k+1}}+\mathbf{Y_2^k}\big)\big(\mu^k\mathbf{I}+2\gamma\mathbf{M}\big). \tag{33}$$

4) *Update* $\mathbf{Y_1^k}$ *and* $\mathbf{Y_2^k}$

$$\begin{aligned}\mathbf{Y_1^{k+1}} &= \arg\min_{\mathbf{Y_1}} \mathcal{L}\big(\mathbf{L^{k+1}},\mathbf{S^{k+1}},\mathbf{Z^{k+1}},\mathbf{Y_1},\mathbf{Y_2^k},\mu^k\big)\\ &= \arg\min_{\mathbf{Y_1}}\left(\frac{(\mu^k)^2}{2}\left\|\mathbf{F}-\mathbf{L^{k+1}}-\mathbf{S^{k+1}}+\frac{\mathbf{Y_1}}{\mu^k}\right\|_F^2 - \frac{1}{2}\|\mathbf{Y_1}\|_F^2\right).\end{aligned} \tag{34}$$

Take the partial derivative for two terms in the above equation, respectively, and we can get the following iterative formula:

$$\mathbf{Y_1^{k+1}} = \mathbf{Y_1^k} + \mu^k\big(\mathbf{F}-\mathbf{L^{k+1}}-\mathbf{S^{k+1}}\big). \tag{35}$$

Similarly, $\mathbf{Y_2^k}$ can be updated by the following equation:

$$\mathbf{Y_2^{k+1}} = \mathbf{Y_2^k} + \mu^k\big(\mathbf{F}-\mathbf{L^{k+1}}-\mathbf{S^{k+1}}\big). \tag{36}$$

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APPENDIX B
DERIVATION OF ALGORITHM 2

In this appendix, we derive (23) from (21). For (21),

$$\mathbf{P} = \arg\min_{\mathbf{P}} \left\{ \frac{1}{2}\|\mathbf{P}\|_F^2 + \frac{\theta}{2} \sum_{i=1}^{N_1+N_2} \tilde{w}_i \left\| \tilde{\mathbf{A}}_\mathbf{i}\mathbf{P} - \tilde{\mathbf{B}}_\mathbf{i} \right\|_F^2 \right\}. \quad (37)$$

Take the partial derivative for the above two terms, respectively

$$\frac{\partial\left(\frac{1}{2}\|\mathbf{P}\|_F^2\right)}{\partial \mathbf{P}} = \mathbf{P} \quad (38)$$

$$\frac{\partial\left(\frac{\theta}{2}\sum_{i=1}^{N_1+N_2} \tilde{w}_i \left\| \tilde{\mathbf{A}}_\mathbf{i}\mathbf{P} - \tilde{\mathbf{B}}_\mathbf{i} \right\|_F^2\right)}{\partial \mathbf{P}} = \theta \tilde{\mathbf{A}}^\mathbf{T}\tilde{\mathbf{W}}(\tilde{\mathbf{A}}\mathbf{P} - \tilde{\mathbf{B}}). \quad (39)$$

Let the above equations equal to zero, then we have the following closed-form solution:

$$\mathbf{P} = \left(\mathbf{I} + \theta \tilde{\mathbf{A}}^\mathbf{T}\tilde{\mathbf{W}}\tilde{\mathbf{A}}\right)^{-1}\left(\lambda \tilde{\mathbf{A}}^\mathbf{T}\tilde{\mathbf{W}}\tilde{\mathbf{B}}\right). \quad (40)$$

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